Kaspian Thomas

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EDUCATION

University of New Orleans | New Orleans, LA **May 2027**

B.S. in Computer Science w/ Concentration in Machine Learning and AI, | GPA: 3.7/4.0

- **Courses:** , Data Struct**,** Systems Programming Concepts**,** Mach Struct & Assembly Lang, Natural Language Processing, Network Telecommunication, Datamodels and Database Systems
- **Honors:** Dean's List x2 (Fall 25' & Spring 26')

RELEVANT EXPERIENCE

F1 Real-Time Telemetry Commentary System — IBM Hackathon **May 2026**

Co-authored · Python · FastAPI · WebSocket · React · IBM Granite · Langflow · SQLite · Netlify

https://github.com/NightsTTV/kast/tree/f1-telemetry-app/f1-ai-commentator/f1-app

- Co-authored a gated **RAG** pipeline that distills ~1,400 lap events per race into ~64 grounded AI commentary moments (~4.5%), using **IBM Granite** (Ollama/Langflow) enriched with 3 years of official FastF1 data (64k+ laps) — reducing LLM overhead by 95% with zero hallucination on race statistics.
- Engineered a real-time async WebSocket server **(FastAPI)** broadcasting live telemetry, standings, and AI commentary to all clients simultaneously; commentary generates as a fire-and-forget task so the 1–3s LLM call never stalls the data stream.
- Deployed end-to-end as a co-author: **React** dashboard live on **Netlify, FastAPI** backend exposed via ngrok static tunnel; architecture migrates from local **IBM Granite** to **IBM watsonx.ai** cloud with zero pipeline changes.

PROJECTS

BioBERT Soccer Injury Severity Classifier **Spring 2026**

Python, PyTorch, HuggingFace Transformers, scikit-learn, Matplotlib/Seaborn, Google Colab (T4 GPU), LaTeX

- Fine-tuned **BioBERT** to classify soccer injury reports into **5 severity tiers** using PyTorch and HuggingFace Transformers, training on Google Colab with a T4 GPU
- Built full evaluation pipeline with **ROC/AUC metrics, confusion matrices, and precision-recall curves** via scikit-learn, matplotlib, and seaborn; compared against random and majority-class baselines
- Authored full academic report in **LaTeX** and delivered a presentation analyzing model performance

Multiplayer Networked UNO Card Game **Spring 2026**

Java, TCP/IP Socket Programming, Multi-threading, ExecutorService, Swing GUI

- Implemented a **multiplayer networked UNO game** in Java enforcing full UNO rules (special cards, draw logic, turn management) over TCP/IP sockets
- Designed a multi-threaded server using **ExecutorService and ConcurrentHashMap** with automated matchmaking, game room management, and countdown timers
- Built a **Swing GUI client** with card hand rendering, game log, draw mechanics, and integrated real-time chat panel

Chatty – Multi-Client Chat Application **Spring 2026**

Java, TCP Sockets, Multi-threading, Swing GUI, ConcurrentHashMap

- Developed a **real-time multi-client chat server** in Java handling concurrent connections via dedicated ClientHandler threads broadcasting to all users over TCP
- Enforced **unique username registration, live user roster, and graceful disconnect** using a thread-safe ConcurrentHashMap; built themed Swing GUI with EDT safety via SwingUtilities

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, SQL, PL/SQL, JavaScript
- **Frameworks/Libraries:** React, Node.js, TensorFlow, PyTorch, HuggingFace Transformers, Pandas, scikit-learn, spaCy, Swing, FastAPI, Pydantic, Uvicorn, Tailwind CSS, Vite
- **Tools:** Docker, Git, PostgreSQL, SQLite, Oracle SQL, Jupyter/Google Colab, LaTeX, Matplotlib, Langflow, Ollama, FastF1, Netlify, ngrok.
- **Methodologies:** Agile, Test-Driven Development, CI/CD, Client-Server Architecture, Algorithm Analysis